

SINOP, Turkey

WMO: 170260

Lat: 42.0297N

Lon: 35.1543E

Elev: 105

StdP: 14.64

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	32.8	34.9	22.1	16.7	39.9	25.3	19.4	40.2	25.9	37.9	23.6	38.8	9.1	10	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	84.1	74.9	82.4	74.2	80.9	73.4	78.1	81.7	76.7	80.4	75.4	79.3	7.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	141.1	80.5	75.4	133.7	79.5	74.1	127.7	78.4	41.7	82.2	40.3	80.6	39.0	79.5	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.7	17.4	14.6	DB	29.6	87.2	3.3	1.4	27.3	88.2	25.4	89.0	23.5	89.8	21.2	90.8	
			WB	27.8	79.0	3.5	2.0	25.3	80.5	23.2	81.7	21.2	82.8	18.7	84.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.1	45.8	45.0	47.2	52.5	61.1	69.4	74.7	76.1	69.6	62.5	54.9	49.1	(6)
(7)		DBStd	12.15	6.13	6.03	5.95	5.82	5.28	3.78	2.88	3.08	4.33	5.75	5.98	6.28	(7)
(8)		HDD50	599	160	162	128	32	0	0	0	0	1	23	93		(8)
(9)		HDD65	3168	595	561	552	378	145	8	0	12	117	308	492		(9)
(10)		CDD50	3910	29	21	41	107	343	583	766	808	589	389	168	66	(10)
(11)		CDD65	1005	0	0	0	3	23	141	301	343	151	40	3	0	(11)
(12)		CDH74	4463	0	0	1	11	44	334	1528	2123	345	72	5	0	(12)
(13)		CDH80	521	0	0	0	1	3	17	152	317	18	13	0	0	(13)
(14)	Wind	WSAvg	6.0	6.4	6.7	7.1	6.4	5.3	5.5	6.2	5.7	5.5	5.5	5.7	6.3	(14)
(15)	Precipitation	PrecAvg	25.6	2.9	1.8	1.6	1.5	1.1	1.2	1.9	2.3	3.4	3.1	3.3		(15)
(16)		PrecMax	39.0	5.8	5.0	3.7	3.1	4.5	3.3	7.2	8.9	5.0	12.8	9.6	6.9	(16)
(17)		PrecMin	13.1	0.1	0.2	0.0	0.3	0.3	0.2	0.0	0.1	0.7	0.5	0.9		(17)
(18)		PrecStd	6.0	1.3	1.0	0.9	0.8	0.9	0.7	1.5	1.9	1.3	2.5	1.9	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.8	64.8	70.2	75.5	78.2	82.3	85.5	87.0	82.4	80.8	73.9	68.4	(19)
(20)		0.4%	MCWB	51.1	51.4	53.9	58.2	66.6	70.5	74.7	77.5	72.9	63.8	56.8	53.8	(20)
(21)		2%	DB	60.9	59.9	62.7	68.8	74.5	79.8	83.3	84.7	79.4	75.4	68.4	63.7	(21)
(22)		2%	MCWB	49.5	49.2	51.4	57.0	64.5	70.9	73.8	76.4	70.8	64.8	57.1	52.7	(22)
(23)		5%	DB	57.7	56.7	58.9	64.5	71.3	77.7	81.7	83.2	77.6	72.5	65.2	60.6	(23)
(24)		5%	MCWB	48.3	48.2	50.0	55.4	63.8	69.5	73.6	75.6	70.0	65.3	57.5	51.4	(24)
(25)		10%	DB	54.4	53.3	55.4	60.4	68.7	75.8	80.3	81.7	75.9	69.9	62.5	57.8	(25)
(26)		10%	MCWB	47.3	47.0	49.1	53.4	62.6	68.9	73.3	74.7	69.1	64.7	56.5	50.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	53.4	55.9	62.7	72.1	76.3	79.6	80.6	75.6	69.7	63.7	56.7	(27)
(28)		0.4%	MCD B	59.9	59.7	64.3	67.2	73.2	78.3	81.9	83.5	80.0	73.2	66.0	62.3	(28)
(29)		2%	WB	51.0	51.3	53.2	58.7	67.6	73.8	77.3	78.9	73.5	68.3	61.2	54.7	(29)
(30)		2%	MCD B	57.8	57.1	59.4	66.6	70.6	76.8	80.8	82.7	76.9	71.7	63.2	59.6	(30)
(31)		5%	WB	49.6	49.5	51.3	56.1	65.4	71.9	76.0	77.4	71.9	67.0	59.3	53.2	(31)
(32)		5%	MCD B	54.8	54.5	56.7	62.8	68.9	75.4	79.7	81.3	75.6	70.4	62.3	57.8	(32)
(33)		10%	WB	48.3	47.9	49.7	54.2	63.5	70.5	74.8	76.3	70.5	65.2	57.4	51.7	(33)
(34)		10%	MCD B	52.3	51.5	54.9	59.0	67.5	74.2	78.7	80.3	74.3	69.0	61.2	56.5	(34)
(35)	Mean Daily Temperature Range		MDBR	7.1	7.9	8.5	8.9	9.3	9.7	9.4	9.1	8.3	8.1	7.5	(35)	
(36)		5% DB	MCD BR	12.3	14.5	15.5	15.7	13.4	12.4	11.1	11.0	10.8	11.5	12.5	11.4	(36)
(37)		5% DB	MCWBR	6.1	7.3	7.5	7.9	8.0	7.4	6.3	6.6	6.2	6.1	6.6	6.3	(37)
(38)		5% WB	MCD BR	10.0	12.1	13.2	14.1	12.2	10.5	9.7	10.0	9.8	9.5	9.6	9.2	(38)
(39)		5% WB	MCWBR	6.3	6.9	7.5	8.3	8.9	8.2	7.0	7.2	6.7	6.2	6.6	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.352	0.384	0.422	0.461	0.434	0.413	0.426	0.435	0.410	0.397	0.362	0.349	(40)	
(41)		taud	2.364	2.265	2.174	2.100	2.265	2.375	2.356	2.319	2.375	2.357	2.398	2.383	(41)	
(42)		Ebn at Noon	246	255	259	257	268	274	268	262	261	248	241	236	(42)	
(43)		Edh at Noon	27	34	42	49	42	38	38	39	34	31	26	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	480	715	1098	1546	1873	2182	2209	1920	1435	930	619	427	(44)	
(45)		RadStd	45	87	79	154	179	141	106	101	109	86	66	49	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)	
(47)	Regional (5 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	-148	-257	+191	+119	(47)

Nomenclature: See separate page